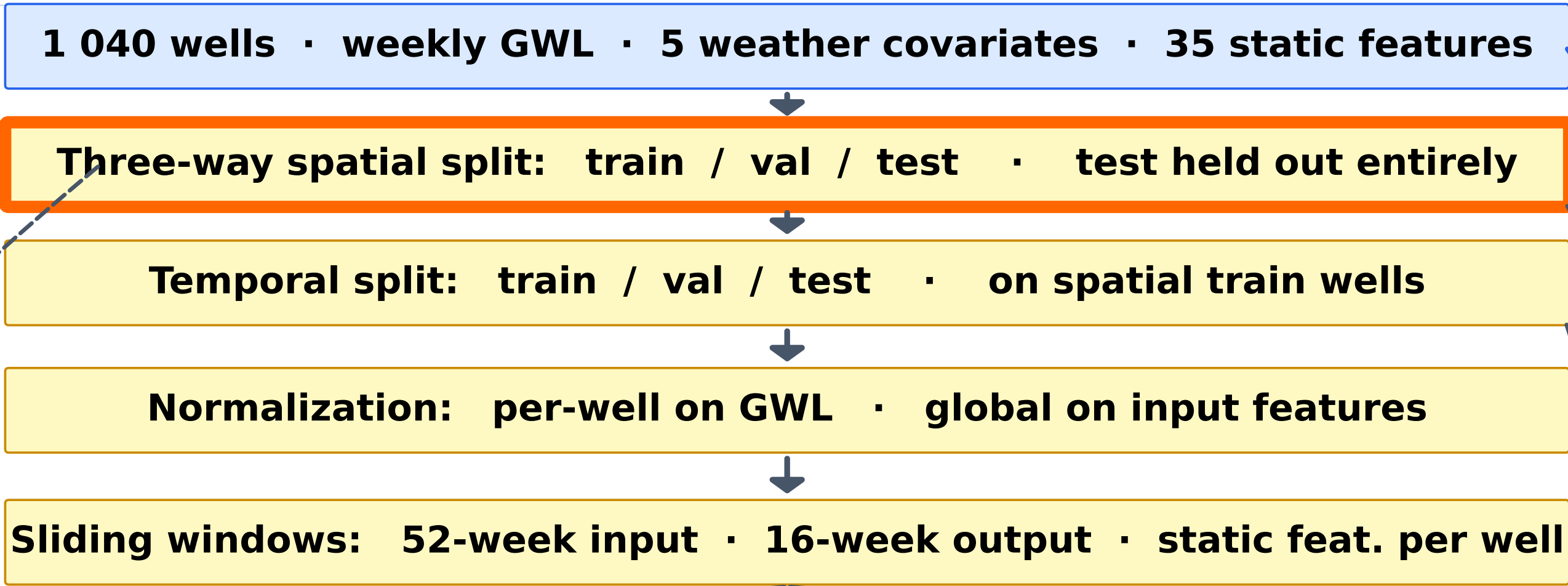


① Data Preparation



Legend

Data

Processing

GRU component

Transferred hidden state

GP component

Loss / objective

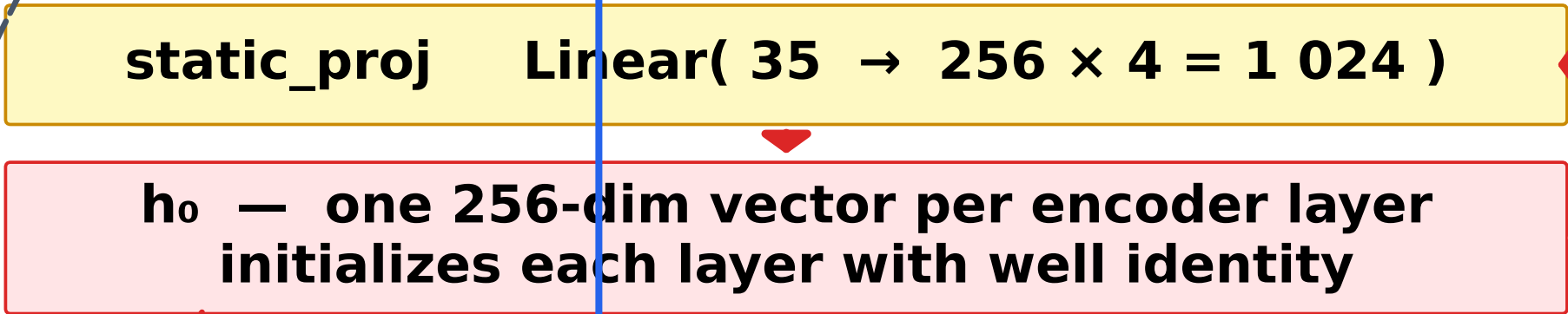
Joint training only

Gradient backprop

② Model Inputs (batch size B)

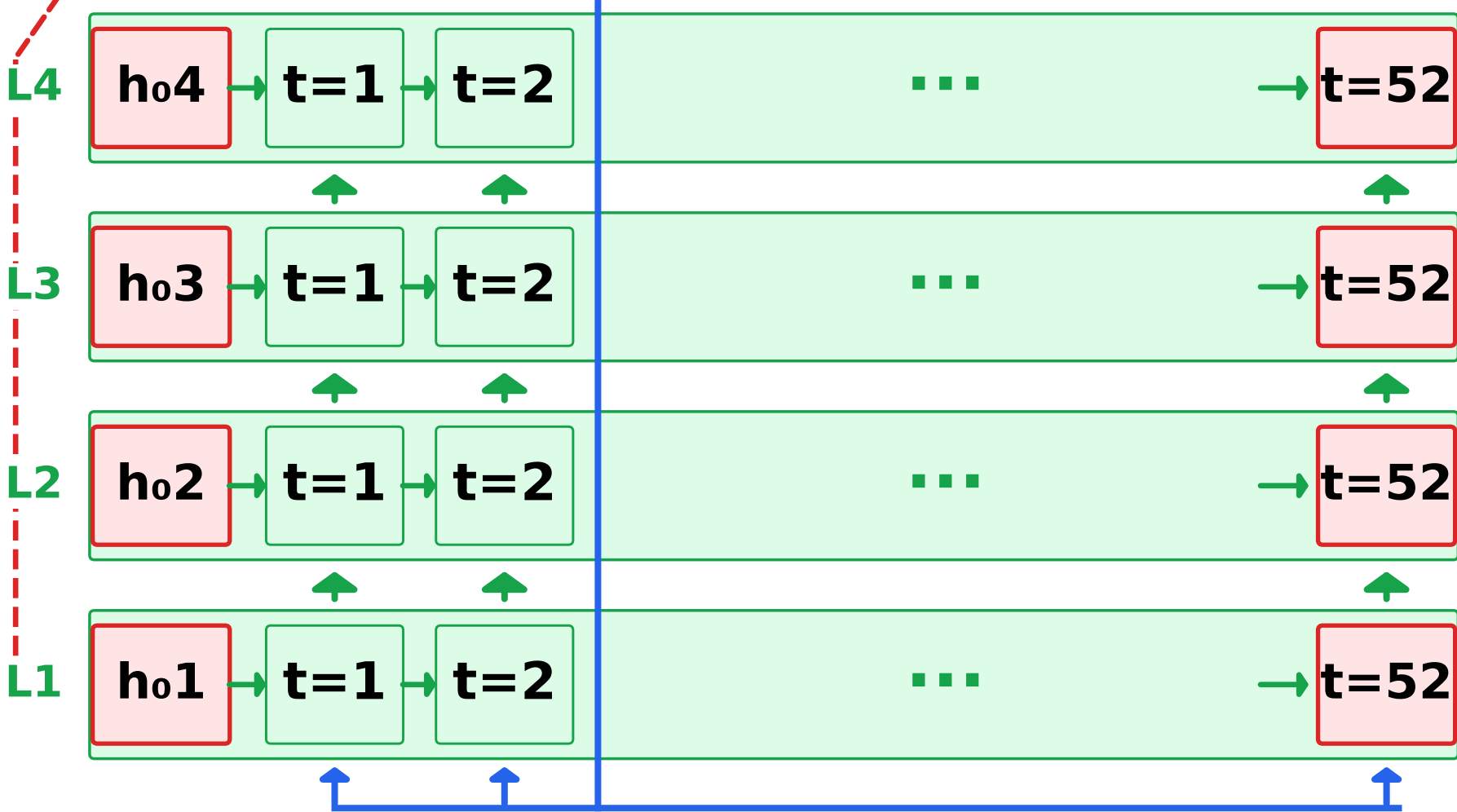


③ Static Projection → Encoder Initial State h_0



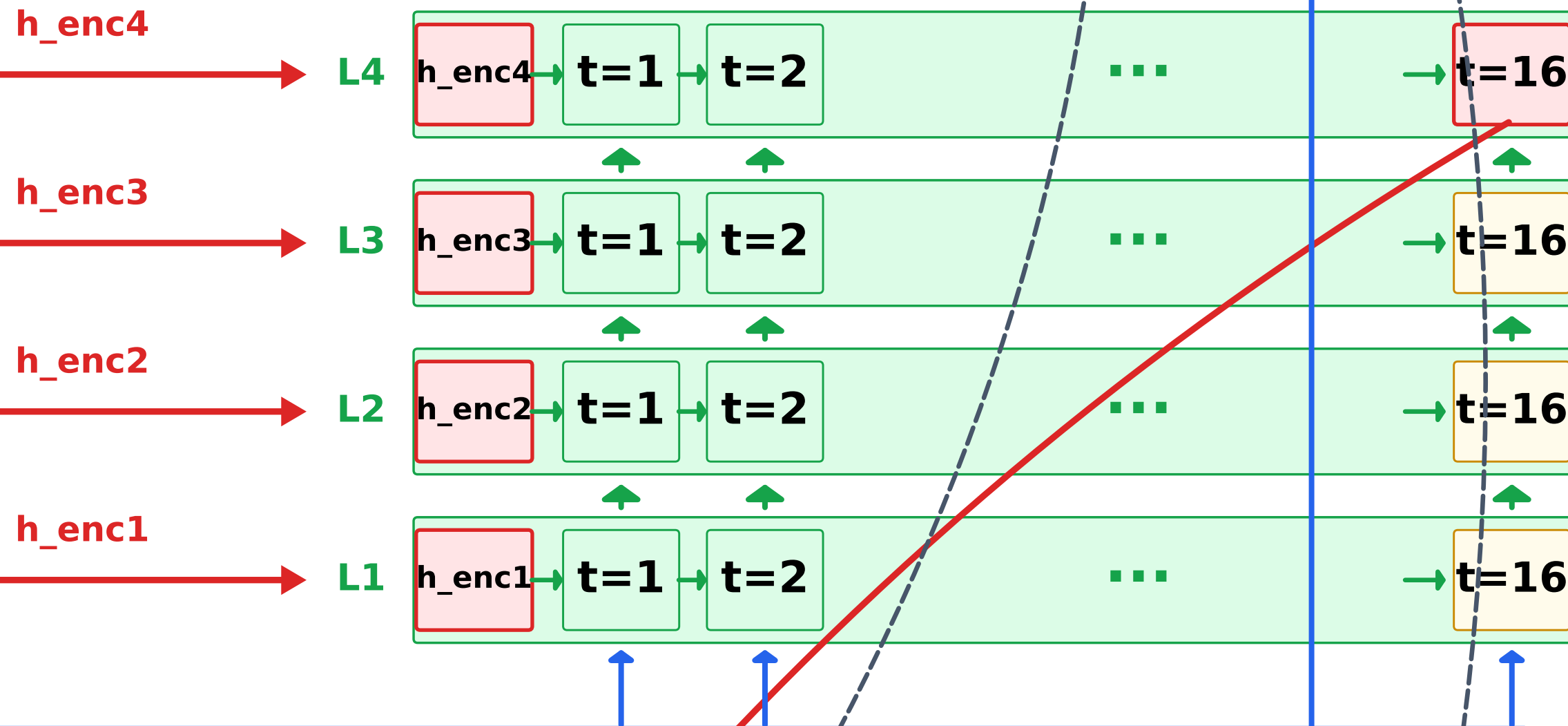
④ Encoder GRU (52 steps)

4 layers × 256 nodes · dropout 0.2

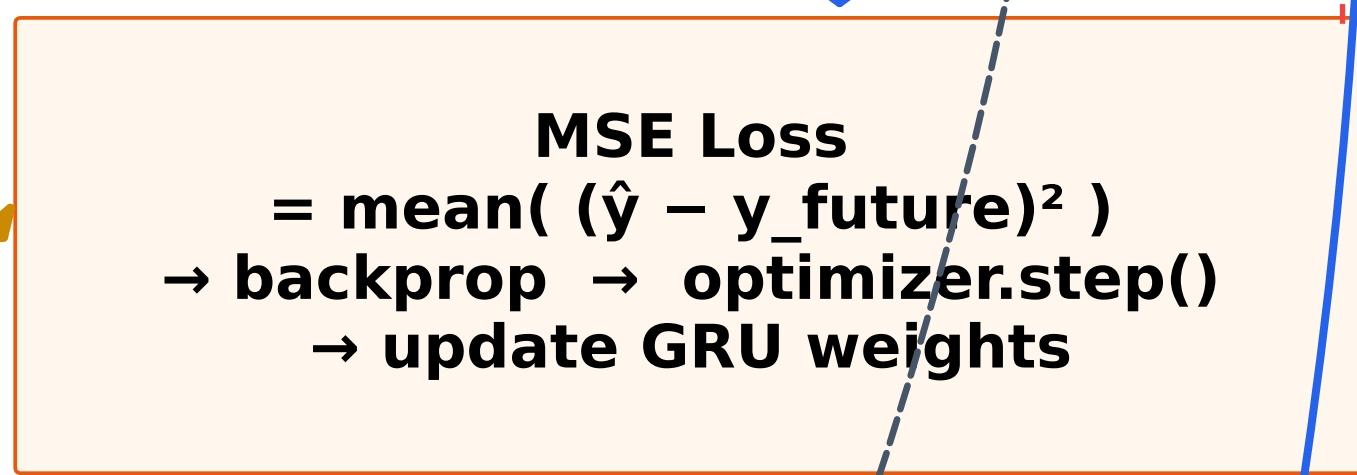
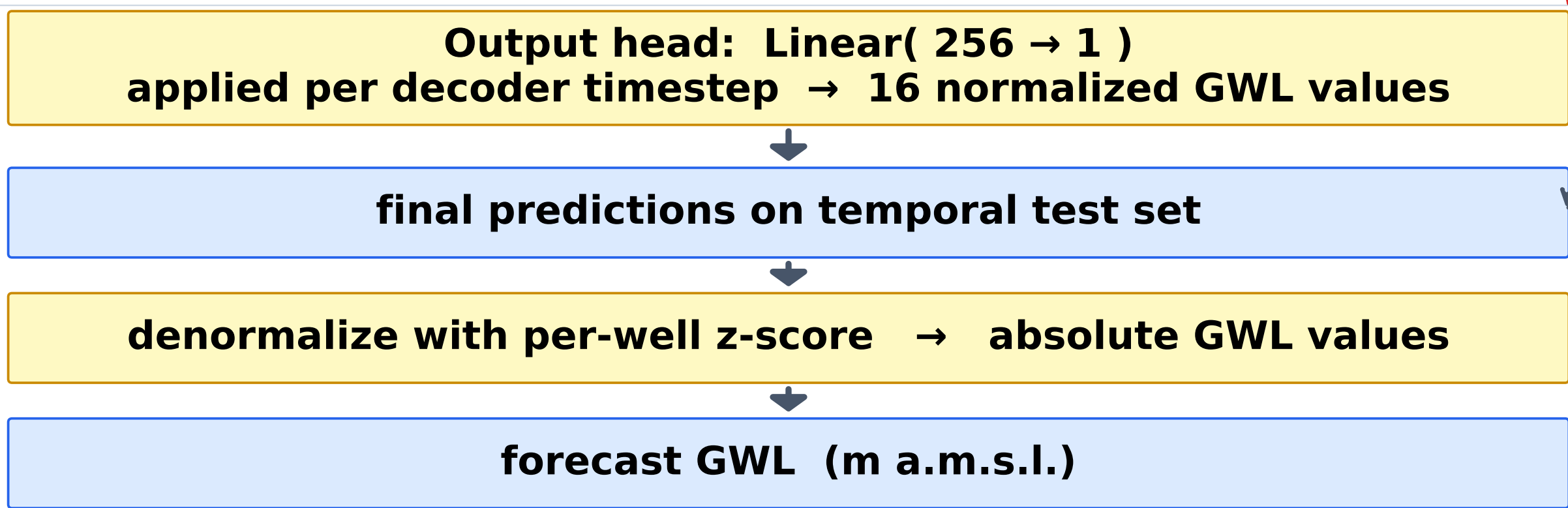


⑤ Decoder GRU (16 steps)

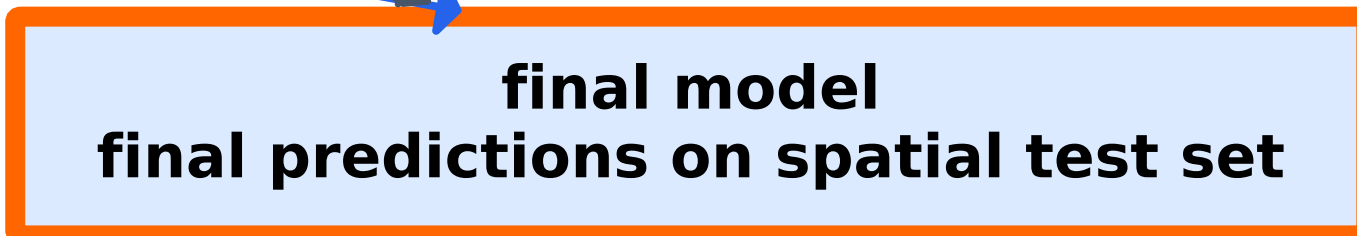
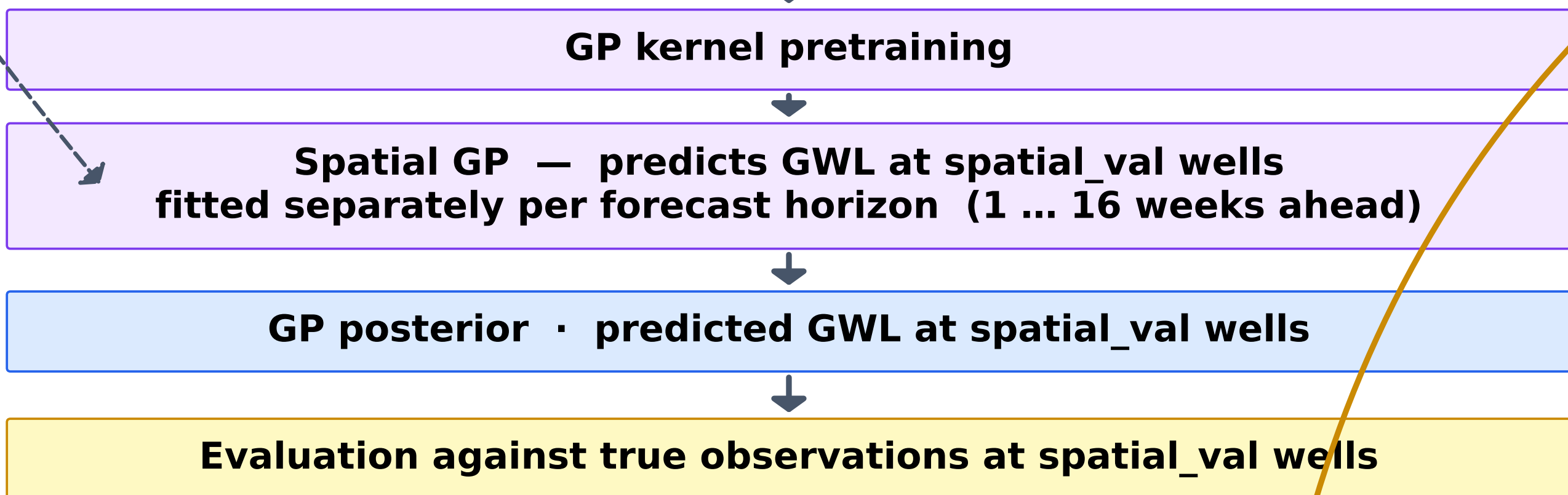
4 layers × 256 nodes · dropout 0.2



⑥ Output Head, Denormalization & Predictions



⑦ GP Spatial Interpolation (spatial_val wells)



⑧ Joint Objective

